

Aalok Thakkar

aalok-thakkar.github.io

Academic Appointment

2024–PRESENT ASHOKA UNIVERSITY, Sonipat, India
Department of Computer Science
Assistant Professor

Education

2018–2023 UNIVERSITY OF PENNSYLVANIA
Ph.D., Computer and Information Science
Thesis: *Example-Guided Synthesis of Relational Queries*
Advisors: Rajeev Alur and Mayur Naik

2015–2018 CHENNAI MATHEMATICAL INSTITUTE
B.Sc. (Honours), Mathematics and Computer Science

Publications

Conference Publications

The Trust a Protocol Needs

with Bhumika Mittal

International Conference on Information Systems Security (ICISS 2026)

Defining Purpose Limited Secrets

with Bhumika Mittal

International Conference on Information Systems Security (ICISS 2026)

Localising Stochasticity in Weighted Automata

with Smayan Agarwal

Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2026)

How Often Does Your Program Fail?

with Arnab Ray

Automated Technology for Verification and Analysis (ATVA 2026)

A Forced-Structure Reduction and Verifiable Bounds for Conway's 99-Graph

Conference For AI Scientists (CAISc 2026) — **Best Paper Award**

A Correctness-First Introduction to Computer Science

with Vaani Goenka

Innovation and Technology in Computer Science Education (ITiCSE 2026)

A Good Rubber Duck Does Not Quack

with Ayush Thonge

Innovation and Technology in Computer Science Education (ITiCSE 2026)

Accessibility Beyond Accommodations

with Vaanee Tripathi

ACM India Compute Conference (COMPUTE 2025)

Ancient Algorithms for Modern Curriculum

ACM India Compute Conference (COMPUTE 2025)

BOOP: Write Right Code

with Vaani Goenka

ACM India Compute Conference (COMPUTE 2025)

SocraticAI: Transforming LLMs into Guided CS Tutors Through Scaffolded Interaction

with Karthik Sunil

ACM India Compute Conference (COMPUTE 2025)

From Transformers to Weighted Automata: Towards Verification of Large Language Models

with Smayan Agarwal, Aslah Ahmad Faizi, and Shobhit Singh

From Data to Models and Back (DATAMOD 2025)

Mobius: Synthesizing Relational Queries with Recursive and Invented Predicates

with Nathaniel Sands, George Petrou, Rajeev Alur, Mayur Naik, and Mukund Raghothaman

Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA 2023)

Relational Query Synthesis $\bowtie$ Decision Tree Learning

with Aaditya Naik, Adam Stein, Rajeev Alur, and Mayur Naik

International Conference on Very Large Data Bases (VLDB 2023)

Example-Guided Synthesis of Relational Queries

with Aaditya Naik, Nathaniel Sands, Mukund Raghothaman, Mayur Naik, and Rajeev Alur

Programming Language Design and Implementation (PLDI 2021)

Additive Collatz Trajectories

with Mrunmay Jagadale

International Conference of The Jangjeon Mathematical Society (ICJMS 2016)

Journal Publications

Reopening businesses and risk of COVID-19 transmission

with Ashley O'Donoghue, Tenzin Dechen, Whitney Pavlova, Michael Boals, Garba Moussa, Manvi Madan, Frank J. DeFalco, and Jennifer P. Stevens

npj Digital Medicine 4, 51 (2021)

Concurrency in Boolean networks

with Thomas Chatain, Stefan Haar, Juraj Kolcak, and Loic Pauleve

Natural Computing 19, 91–109 (2020)

Infinitude of Primes Using Formal Languages

The American Mathematical Monthly 125:8, 745–749 (2018)

*Workshop Publications***Complexity of Relational Query Synthesis**

with Rajeev Alur and Mayur Naik

*11th Workshop on Synthesis (SYNT 2022)***Modular Synthesis of Reactive Systems**

with Kedar S. Namjoshi and Richard Treffer

*9th Workshop on Synthesis (SYNT 2020)***Verification of Rewriting-based Query Optimizers**

with Krishnamurthy Balaji and Piyush Gupta

*16th International Workshop on Termination (WST 2018)***Thesis**

2023	Example-Guided Synthesis of Relational Queries Ph.D. thesis, University of Pennsylvania.
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Teaching

SUMMER 2026	FC-0306: Quantitative Reasoning and Mathematical Thinking
SPRING 2026	CS-3330: Theory of Computation CS-3610: Information Security
MONSOON 2025	CS-1102: Introduction to Computer Science CS-5310: Symbolic Logic and Applications CS-6101: Computer Science Research Methods and Seminars
SUMMER 2025	FC-0306: Quantitative Reasoning and Mathematical Thinking CS-1102: Introduction to Computer Science
SPRING 2025	CS-1102: Introduction to Computer Science CS-5440: Trustworthy AI
MONSOON 2024	CS-5310: Symbolic Logic and Applications

Grants and Fellowships

2026–2027	INSA VISITING SCIENTIST PROGRAMME Visiting the Chennai Mathematical Institute to work with Aiswarya on the synthesis of string transducers.
2026–	ANRF PRIME MINISTER EARLY CAREER RESEARCH GRANT Robust and scalable program synthesis.

2025–2026	DATA & AI ADOPTION FELLOWSHIP, EkStep Foundation AI Safety Benchmarks for Education and Agriculture.
2025–2026	DARJEELING REVIVAL FELLOWSHIP, Alt Carbon Geostatistical modelling for enhanced rock weathering.
2025–2026	ADAPTIVE AND INCLUSIVE PEDAGOGY GRANT, Mphasis Curriculum design for an adaptive and inclusive classroom.
2026	NAGARATHNA MEMORIAL GRANT Translation of <i>Nayi Kavita</i> poets into English.

Professional Service

ORGANIZER	Formal Methods Update Meeting 2026, Krea University. FSTTCS 2024, FSTTCS 2025. IARCS Verification Seminar Series. Logic Society of Delhi.
EDITOR	Corresponding editor, <i>Bhāvanā</i> magazine.
REVIEWER	Programme committees and review activities for POPL AEC 2022, ISEC 2025, PLDI SRC 2025, ATVA AEC 2025, OOPSLA 2026, and <i>ACM Computing Surveys</i> .
MEMBERSHIPS	Association for Computing Machinery (ACM India). Vijnana Bharati.